The technological basis for adaptive ion beam therapy at MedAustron: Status and outlook

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Abstract

The ratio of patients who need a treatment adaptation due to anatomical variations at least once during the treatment course is significantly higher in light ion beam therapy (LIBT) than in photon therapy. The ballistic behaviour of ion beams makes them more sensitive to changes. Hence, the delivery of LIBT has always been supported by state of art image guidance. On the contrary CBCT technology was adapted for LIBT quite late. Adaptive concepts are being implemented more frequently in photon therapy and also efficient workflows are needed for LIBT. The MedAustron Ion Beam Therapy Centre was designed to allow the clinical implementation of adaptive image-guided concepts. The aim of this paper is to describe the current status and the potential future use of the technology installed at MedAustron. Specifically addressed is the beam delivery system, the patient alignment system, the treatment planning system as well as the Record & Verify system. Finally, an outlook is given on how high quality X-ray imaging, MR image guidance, fast and automated treatment planning as well as in vivo range verification methods could be integrated.

Keywords: Light ion beam therapy, Radiotherapy, Adaptive

1 Introduction

The physical characteristics of light ion beams and the resultant better conformity and dosimetric advantage over high-energy photon beams may lead to reduced toxicity and potentially provides the possibility for dose escalation to the tumour. Besides the ballistic selectivity there are biological advantages that depend on the used particle species. These physical and biological benefits can only be translated into clinically relevant advantages if treatment planning and dose delivery is based on state-of-the art technology for imaging, i.e. imaging for target definition and daily image guidance in- or outside the treatment room [1–3].

CT-based treatment planning was pioneered in the mid-70s by the Loma Linda University Medical Centre in California for the purpose of proton therapy and in-room image guidance for proton therapy was ahead of photon therapy in the 70s, 80s and 90s [4]. The picture in terms of advanced image guidance

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is different in today's radiotherapy setting. The leading role in image guidance in proton therapy several decades ago can be explained by the fact that the proton beam itself could not be used for imaging and consequently X-ray devices were applied for daily image guidance. Furthermore, the risk of missing the target due to the sharp distal dose fall off in combination with range uncertainties motivated daily set-up and anatomy verifications. Since the 1960s, the traditional method of aligning patients at a LIBT facility was based on orthogonal pairs of kV X-ray images that were obtained every fraction in the treatment room [4]. The technique generally used the patient's bony anatomy as alignment reference due to its relationship to the target volume and the critical tissues. More recently, also cone-beam CT (CBCT) technology has become available for image guidance in LIBT delivery [1,5]. Provided sufficient image quality is achieved, the soft tissue information can be of further value for image guidance of patients with tumours, which are not directly linked to bony anatomy [6].

During the last decade scanned ion beam delivery superseded passive scattering for LIBT. This further motivated image guidance in LIBT due to the pronounced sensitivity of the particle range to anatomical changes or setup errors [7]. However, this also leads to the dilemma that treatment adaptations due to anatomical changes are much more frequently needed than in photon therapy. Due to the anatomical changes, it is also difficult to track the actual delivered dose throughout the course of therapy. It is estimated that roughly one third of all LIBT patients require at least one re-planning during the course of their treatment. A process, which in LIBT still implies a lot of additional workload. Often the entire planning process from planning CT to patient specific QA needs to be repeated. Compared to the passive beam delivery, computerised treatment plan optimisation in scanned LIBT does not need to fabricate new passive elements, but re-planning requires comprehensive patient specific QA.

Adaptive concepts are being implemented more frequently in photon therapy [8] with still some additional workload. Highly innovative technology is therefore needed to allow faster and more precise treatment adaptations in the context of LIBT while reducing the overall workload. Within this manuscript, it is aimed to address the current status of the MedAustron LIBT centre and to highlight parts of its recent and future technology as well as visions to implement fast and precise treatment adaptation techniques.

2 Overview and current status of the technology installed at MedAustron

The MedAustron facility is a dual particle facility for proton and carbon ion beam therapy, which started its clinical operation with proton beams in one treatment room (see Fig. 1) as well as its research operation in a dedicated non-clinical research room in 2016. The research room is equipped with the same patient positioning and imaging system as the medical treatment rooms. The clinical operation with protons in a second treatment room started in 2017. The medical technology available at MedAustron was designed from scratch to enable adaptive LIBT.

Figure 1. Birds eye view of the MedAustron facility. Currently the treatment room number two and three with a horizontal fixed beam line are in clinical operation with protons. The research room is also in operation with proton beams that have settings similar to the clinical room. The other room with the proton gantry and carbon ions will be clinically commissioned in the forthcoming months and years.

The MedAustron Particle Therapy System (MAPTS) consists of several components and medical products. One of the main components is the MedAustron Particle Therapy Accelerator (MAPTA), which is a synchrotron-based accelerator system providing beam in all treatment and research rooms. A so-called Medical Front-end part ensures the safe application of the ion beams in all rooms. The treatment nozzle in each irradiation room comprises passive beam modifiers, i.e. a range shifter and ripple filters, which can be automatically moved in and out of the beamline according to the prescribed treatment plan.

The unique and newly designed patient alignment system (PAS) includes the robotic patient positioning system (Exacure, BEC GmbH, Reutlingen, Germany) and the X-ray based verification system (ImagingRing^{TM}, medPhoton GmbH, Salzburg, Austria). For treatment planning a multi-slice CT scanner (Brilliance CT Big Bore Oncology, Philips, Eindhoven, The Netherlands) was commissioned and structure definition is in addition supported by a high-end 3T MR scanner (Ingenia, Philips, Eindhoven, The Netherlands). Next to the treatment rooms a PET-CT (Gemini TF Big Bore) was installed to serve two purposes, first as backup CT and second to enable research for in vivo range verification.

The treatment planning system RayStation (RaySearch Laboratories, Stockholm, Sweden) was customized and fully integrated to support the clinical workflow. Finally, the whole treatment workflow inside the treatment room is supported by a specifically developed record and verify system in

collaboration with the radART institute (Paracelsus Medical University, Salzburg).

All these devices are CE certified independently from each other and are described in more details below in terms of their functionality and use for adaptive radiotherapy. Specific compatibility tests ensured the correct clinical functionality beyond the interfaces of the medical products.

2.1 Beam delivery system

The accelerator is based on the so called PIMMS (Proton-Ion Medical Machine Study) design, similar to the accelerator at the CNAO facility in Pavia, Italy [9,10]. By design, the synchrotron is able to accelerate charged particles and to provide several particle species for therapy or research, covering the range from protons up to neon ions. The extraction mechanism is Betatron driven to facilitate a slow extraction and a quasi-discrete spot scanning technique. Quasi-discrete scanning is a variant of discrete scanning in which the ion source is not turned off during the move from one pencil beam position to the next [11].

The clinically utilized energies for protons are ranging from 62.4 MeV up to 252.7 MeV. This translates into ranges in water from 30 mm up to 380 mm. Ranges lower than 30 mm can be achieved by inserting the range shifter. The specified step size for proton ranges (R_{80}) in water is 1 mm between 30 and 188 mm, and 2 mm beyond 188 mm, with a total of 255 energy steps. During acceptance and commissioning all proton ranges were verified with a water phantom (MP3 PL from PTW, Freiburg, Germany) and found to be within 0.3 mm of specifications. For research purposes, protons can be accelerated up to 800 MeV and be delivered in the dedicated research room.

For carbon ions the specified energies vary from 120 to 400 MeV/u with ranges from 30 to 280 mm in water with a step size of 1 mm. The technical commissioning for carbon ions started and is currently ongoing.

Nominal spot sizes for both protons and carbon ions range from 4 mm to 10 mm full width half maximum (FWHM) at isocentre in vacuum. During acceptance and commissioning proton spot sizes and positions were verified using a scintillator screen (Lynx detector from IBA dosimetry, Schwarzenbruck, Germany). Delivered spot positions were found to be within 0.5 mm of the prescribed position. The symmetry of the spot size was within specifications (10% or 1 mm, whichever is smaller). The spot size in air at isocentre varies from 21.4 mm FWHM for 62.4 MeV to 7.4 mm FWHM for 252.7 MeV. The spill duration is adjustable from 1 s to 10 s (gating interface to interrupt the beam is still in the design phase). The accelerator can operate with 4 different nominal intensities namely 0%, 50%, 80% and 90% degradation. 0% degradation corresponds to 2×10^{10} particles per spill for protons and 4×10^{8} for carbon ions later on. To reduce the medical commissioning workload it was decided to start clinical operation with protons with a single spill duration of 5 s, one spot size (the smallest) and two intensities (degradation of 80% and 90%).

Each beam line contains a nozzle equipped with the Dose Delivery System developed at CNAO, which is a redundant beam monitoring system. Each beam monitor consists of a pair of strip chambers to monitor the spot sizes and positions (1.65 mm resolution) and an integral chamber to monitor the number of particles delivered [12]. A third monitor is acting as independent termination system. In addition, the automatically moveable passive beam modifiers are situated as close as possible to the nozzle exit. A 30 mm thick range shifter of PMMA and two ripple filters with a thickness of 2 mm each can be inserted independently [13]. A schematic illustration is shown in Fig. 2. The accelerator and nozzle designs allow for a maximum field size of 20 cm × 20 cm at isocentre for the fixed beam lines and 12 cm × 20 cm for the proton gantry.

2.2 Patient alignment system (PAS)

The overall goal of the patient alignment system (PAS) is to accurately position the target within the immobilized patient with respect to a treatment beam as defined during the treatment planning process. The PAS comprises a ceiling mounted robot including a couch-mounted imaging system called the ImagingRing™ (see Fig. 3). The major components of the PAS are: the robotic patient positioning system (PPS), the patient position verification system (PPVS) with an X-ray source and control as well as comprehensive image processing and reconstruction software. For a save patient alignment both systems are equipped with a software collision avoidance system and an active collision detection system (touch guards). The first component is used to set up the immobilized patient in the irradiation room in treatment position, while the second component is used to verify that the immobilized patient is set up correctly as intended in the underlying treatment plan.

2.2.1 Robotic patient positioning system

The robotic PPS consists of a six axes KUKA robot (Augsburg, Germany) that is mounted on an additional linear axes on the ceiling. These seven degrees of freedom guarantee a high flexibility of patient positioning in front of the fixed nozzle. Commissioning was performed for 3° pitch and 3° roll to allow for patient setup correction. In order to assure high positioning accuracy a weight dependent calibration of the system is commonly performed in other facilities [14]. To be independent of the payload, a different solution was implemented at MedAustron, which is based on a floor-mounted photogrammetric tracking camera and reflectors on the bottom side of the table-top. The tracking camera guarantees payload-independent position accuracy by using the optical measurement signals in a feedback loop of the control software for the robot (see Fig. 4).

During the medical commissioning phase the overall accuracy in the treatment volume, which has a size of 115 cm

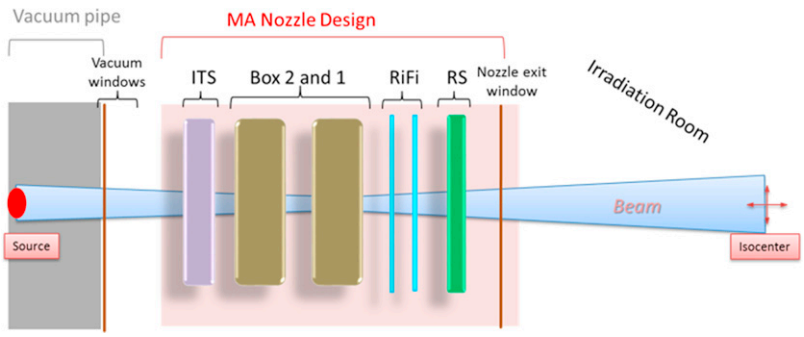

Figure 2. Schematic of the nozzle for a fixed beam line at MedAustron (MA). It shows the position of the two monitor boxes of the dose delivery system (DDS) as well as the independent termination system (ITS) and the passive elements like ripple filter (RiFi) and range shifter (RS).

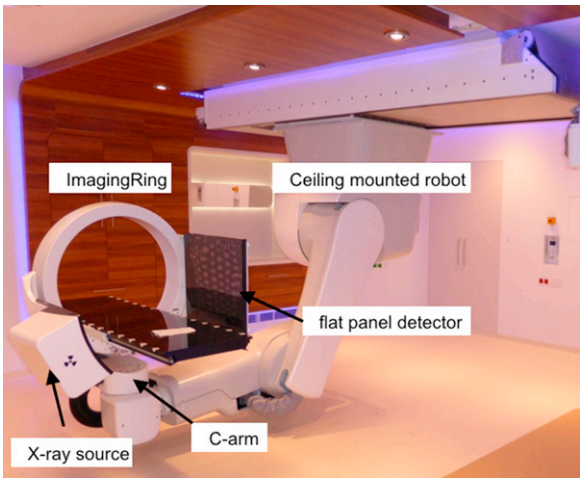

Figure 3. Patient alignment system (PAS) in the treatment room: combined ceiling mounted patient positioning system (PPS) and X-ray patient position verification system (PPVS).

Table 1
Weight dependent position accuracy of the patient positioning system. In total 1020 different positions were measured.

Number of setups	0 kg 475	150 kg 472	Overall 1020
X (in beamline)			
Mean	0.25 mm	0.25 mm	0.24 mm
95% conf. interv.	0.38 mm	0.40 mm	0.40 mm
Y (horizontal)			
Mean	−0.03 mm	0.01 mm	−0.22 mm
95% conf. interv.	0.49 mm	0.44 mm	0.45 mm
Z (vertical)			
Mean	−0.23 mm	−0.21 mm	−0.22 mm
95% conf. interv.	1.09 mm	1.14 mm	1.09 mm
3D			
Mean	0.44 mm	0.43 mm	0.43 mm
0.95 quantile	0.62 mm	0.64 mm	0.63 mm

(length), 60 cm (width), 40 cm (height), was benchmarked with a laser tracker (Leica, Germany) (see Table 1). The position accuracy was found to be comparable to other centers [14] although the maximum possible treatment volume itself is bigger than in other centers and amounts to about 205 l. Independent of the payload, the mean deviation in 3D

Figure 4. The setup of the PPS with tracking camera and laser tracker in the treatment room is shown on the left. It mainly shows the setup during commissioning. The pattern of reflectors on the bottom side of the treatment couch is illustrated on the right.

was found to be smaller than 0.5 mm during commissioning [15,16].

2.2.2 Couch mounted X-ray system

The novel image guidance device is installed in all irradiation rooms at MedAustron (including the research room). The ImagingRing™ system is integrated with a table-top, which is fixed on a C-arm and this is attached to the robotic PPS. The main parts of the ImagingRing™ are a dynamically 2D-collimated X-ray mono block system and a solid-state amorphous silicon detector, which are in turn mounted on the ImagingRing™ gantry support arms. The ImagingRing™ allows imaging of all relevant anatomical regions via longitudinal movement of the ring and independent rotation of the supporting arms. This flexibility allows imaging of patient specific, off-centered (non-isocentric) regions of interest in the transversal plane.

Due to independently moveable source and detector support arms, a transversal field of view (FOV) of up to 700 mm (diameter) and extended planar imaging are achievable with the ImagingRing™. Furthermore, CBCT acquisitions including full (360° arc) and short (180 + fan-angle°) scans as well as region of interest (ROI) imaging with dynamic collimation are possible. ROI imaging reduces the dose to the patient and at the same time increases the image quality as also X-ray head scatter radiation will be reduced. The X-ray source is equipped with 4 independently moveable jaws to be able to limit the X-ray field two dimensionally to the detector area or subregions of it for all constellations of detector to source angles. The X-ray source consists of a mono-block and allows for dual energy imaging to enable soft-tissue tracking. An overview about the ImagingRing™ technical specification is provided in Table 2. The geometrical relationship of the ImagingRing™ to the treatment beam is determined by a two-stage calibration. First, the ImagingRing™ generates X-rays and CBCTs with respect to the table top where the patient is positioned and the imaging device is mounted using an intrinsic flexmap calibration [5,17,15]. Second, the optical tracking camera of the patient positioning system is measuring and correcting the position and orientation of the table top in room coordinates and, thus, establishing the relationship of imaging and the treatment isocenter [16]. The geometrical accuracy of the overall calibration pipeline was validated for all possible treatment positions within the imaging volume using an external laser tracking system. The daily quality assurance includes cross checks of those calibrations using a dedicated phantom [5] and also includes a full image guidance workflow and check with room lasers.

Commissioned for clinical use are currently planar imaging with the recent addition of the CBCT functionality for entities ranging from regular cranial FoV to large pelvic FoV imaging. Functionalities, which will be commissioned in the near future are fluoroscopy, image stitching (panoramic imaging) and dual-energy imaging.

Table 2
Technical specifications of the ImagingRing™ System.

	medPhoton IRS (77 cm)
Source axis distance (SAD) (cm)	63
Detector axis distance (DAD) (cm)	41
Source detector distance (SDD) (cm)	104 (opposing)
Acquisition duration (s)	35
Geometrical accuracy	<0.5 mm
Field of view (lateral SFOV–LFOV (ULFOV) × longitudinal)	25–48 (70) × 29 cm
Panoramic field of view	63 cm
Limited field of view	limited to ROI
Spatial resolution	>17 LP/cm
MTF (50%)	10.8
MTF (10%)	17.2
Uniformity	±20 HU
Patient dose	Down to 1 mGy
Clearance	77 cm
Available energy (kV)	40–120 kV
Dual energy	pulse-to-pulse
Trajectory (degrees)	476/482
Half value layer (mm@120 kV)	6.3 mm
Additional filtration	Al, Cu, CuAg
Panel size (cm^2)	40.96 × 40.96
Flat panel resolution	1024 × 1024@400 μm
Gravitational flex correction	9 + 2 DOF
Gantry-mounted/slow movements	No (independent)

The current installation of the ImagingRing™ system enables improved image guidance compared to other imaging solutions in LIBT and supports offline treatment adaptations. This is illustrated with a clinical case of a nasal cavity treatment. Such indications always imply the risk to get very inhomogeneous dose distributions as cavity filling changes may lead into range over or under shoot (see Fig. 5). It would be therefore beneficial to know the status of the cavities on a daily basis.

The high quality planar X-ray imaging system (see Fig. 6) is able to support, and partly substitute high dose 3D CT imaging for the surveillance of the cavity filling status during treatment with a significantly reduced imaging dose contribution. The cavity status can therefore be easily monitored without the need to perform weekly control CTs, which would mean an additional burden for the patient and also additional resources for CT imaging. As the planar images are performed on a daily basis for patient position verification prior treatment the cavity status information is readily available at a much higher frequency compared to weekly CT or MR imaging. If needed, a new planning CT can be triggered in case of a persistent filling status change compared to the initial planning CT. Potentially, treatment plans with and without filled cavities can be generated at the start of the treatment by density override during planning. Based on the daily situation of the cavities, a specific treatment plan can be chosen. Hence, a plan of the day concept based on planar imaging could be implemented and range uncertainties can be reduced [6].

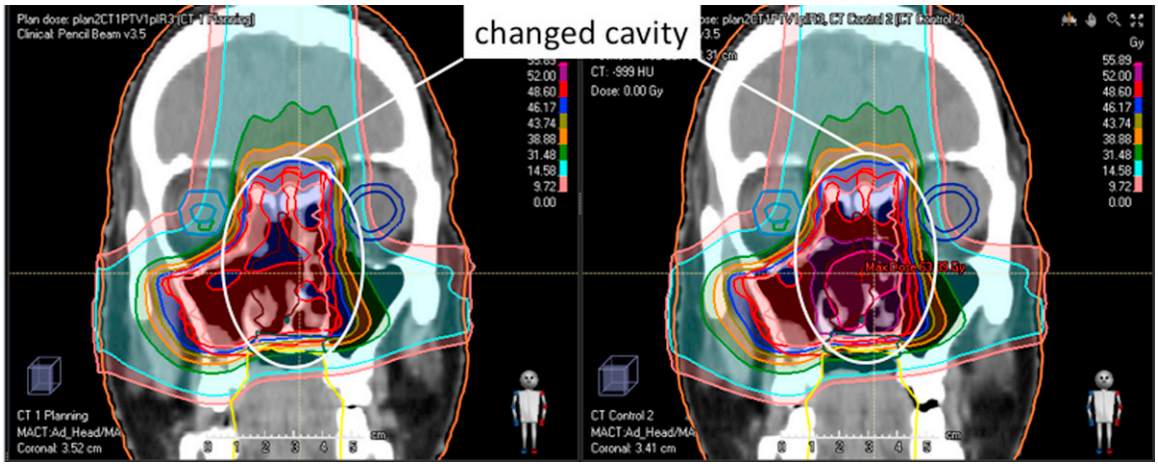

Figure 5. On the left the dose distribution is shown for the initial planning CT. On the right the same plan was recalculated on a new planning CT which was made on the 7th day of treatment. Due to the range overshoot, heterogeneities in the dose distribution are visible.

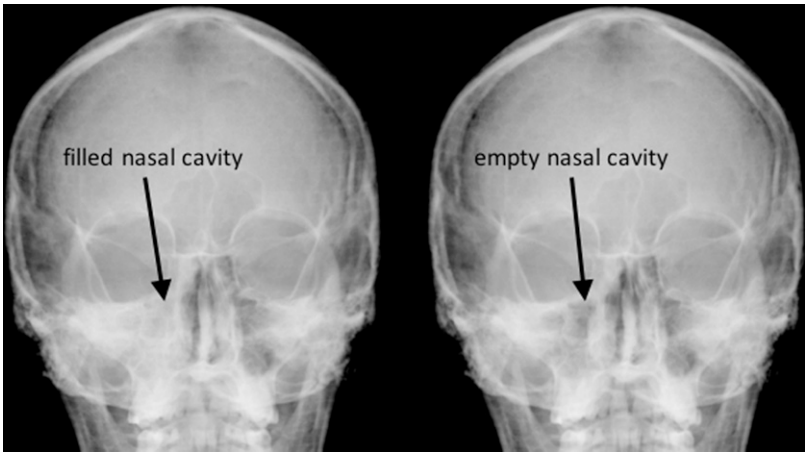

Figure 6. Frontal image (acquired with ImagingRing™) of a patient treated at the nasal cavity. The left image shows the nasal cavity filling at the first treatment day. The right image shows the same nasal cavity already empty after 7 days of treatment similar to Figure 5 before. Following image settings were used: 80 kV, 25 mA, 18 ms, averaging over 5 frames.

Prostate patients treated at MedAustron receive four very thin markers (GoldAnchor, Stockholm, Sweden) that are implanted under ultrasound guidance. Furthermore, the rectum is filled with 100 ml of saline solution [18]. In addition, patients have to follow a specific drinking protocol, i.e. the bladder is emptied half an hour before treatment and then the patients have to drink 500 ml of water to establish a comfortably filled bladder during treatment. Although the patients are encouraged to follow this drinking protocol, it is not guaranteed that the bladder filling can be easily reproduced [6]. This is known to be also very patient dependent. The excellent image quality of the X-ray system allows for bladder visualization, as shown in Fig. 7. Again, those images generated for the patient position verification prior treatment can be used to retrieve information of the bladder filling status. In case the bladder is not full enough, the treatment might be started several minutes later to achieve a similar bladder volume as was the case in the planning CT. In case of a too full bladder, the patient might be asked to void.

Visualization of soft tissue changes on planar images can be realized in two distinct ways depending on the anatomical region. The first approach would be to use the X-ray image from the first fraction as reference and perform additionally a CBCT. Via the CBCT to the planning CT comparison it can be assured that the situation is similar. Further on the reference X-ray can be compared to the subsequent X-ray images from every fraction (preferred scenario for example for filled air cavities). The second approach could be to compare anatomical information from the X-rays with projected organ outlines from the plan structure set (preferred scenario for bladder filling status) or with contrast-optimized digitally reconstructed radiographs derived from the planning CT.

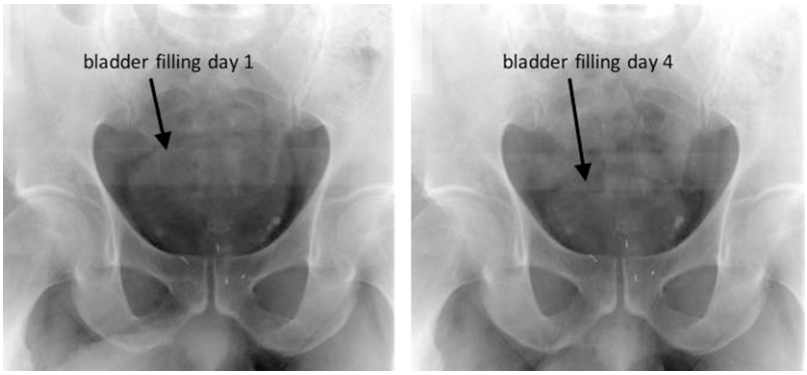

Figure 7. Frontal image of a prostate patient. On the left the bladder status on the first day of treatment is shown, which is comparable to the planning situation. On the right image the bladder status on the fourth treatment is shown. There, the bladder is less filled. Following image settings were used: 120 kV, 25 mA, 18 ms, averaging over 8 frames.

In the past, in-room CT or CBCT technology has been rarely available at most LIBT facilities. Today, more and more LIBT centers aim for CBCT, which is well established in precision photon beam therapy [19]. The main aim is to gain additional volumetric information and to visualize soft tissues, which are barely possible to be visualized with planar X-ray imaging. CBCT technology suffers from variable CT numbers because of the impact of different scatter contributions due to variable patient size or beam hardening and detector-associated scatter glare effects and thus complicates dose re-calculation or re-planning. In addition, due to the relatively slow mode of acquisition, a CBCT is more prone to motion artefacts. Again, this is more relevant for proton therapy compared to photon radiotherapy. More recently, efforts are conducted to improve the CBCT technology [5] and hence make the technology usable for treatment adaptation in the context of LIBT. Preliminary results have been recently published on the possibilities to use a scatter-corrected CBCT for proton dose calculation [20–22]. An example of application of CBCT technology available at MedAustron is shown in Fig. 8.

Imaging dose is still of concern in the context of IGRT [19,23]. Especially in case of LIBT, where daily image guidance is performed. Therefore similar to the method mentioned in Stock et al. [23] we measured the imaging dose for our clinical imaging protocols with TLDs in an anthropomorphic head phantom as well as in an anthropomorphic pelvis phantom. The imaging dose for multi-frame planar imaging was found to be smaller than 0.3 mGy per image in the head region and smaller than 0.8 mGy in the pelvic region. Currently used CBCT protocols produce less than 18 mGy additional dose for the pelvic region and less than 8 mGy in the head region. Therefore, results are comparable to what has been reported for IGRT equipment [19,23]. Furthermore, the imaging dose can in the future be further reduced by incorporating the ROI CBCT imaging capabilities whenever suitable.

2.3 Record and verify system

The Record & Verify system RTS^2 (Radiation Therapy Software System) was developed in the context of a research collaboration by the radART Institute at the Paracelsus Medical University in Salzburg [24]. It was designed as a comprehensive workflow management software. Moreover, literally all in-room sub-systems involved during patient treatment are orchestrated via RTS^2. This includes the beam delivery system components, path-planning for and control of the patient alignment system including offline and online collision checking, 2D and 3D image registration and processing as well as the integration with the irradiation room safety system. The workflow steps are subdivided in so called treatment operations (TOs) and can be planned for every patient in advance. This concept was designed to facilitate adaptive treatment strategies later on. A screenshot including workflow steps is shown in Fig. 9.

2.4 Treatment planning

At MedAustron, the treatment planning system (TPS) RayStation v5.0.2 was accepted for clinical use and the pencil beam algorithm V3.5 was dosimetrically commissioned. The above-mentioned beam delivery characteristics were modelled and validated for both isocentric and also non-isocentric conditions. For fixed beam lines, non-isocentric treatment conditions allows compensating for the non-moveable snout by reducing the air gap between the patient and the nozzle and hence to minimize the beam widening for protons. In addition, the dose calculation accuracy for analytical algorithms is compromised if the combination of a large air gap and a range shifter is used and Monte Carlo algorithms are recommended. Therefore, special attention was paid to non-isocentric setups. Fig. 10 underlines the good relative agreement between a spot profile using a range shifter at a non-isocentric setup. On top of

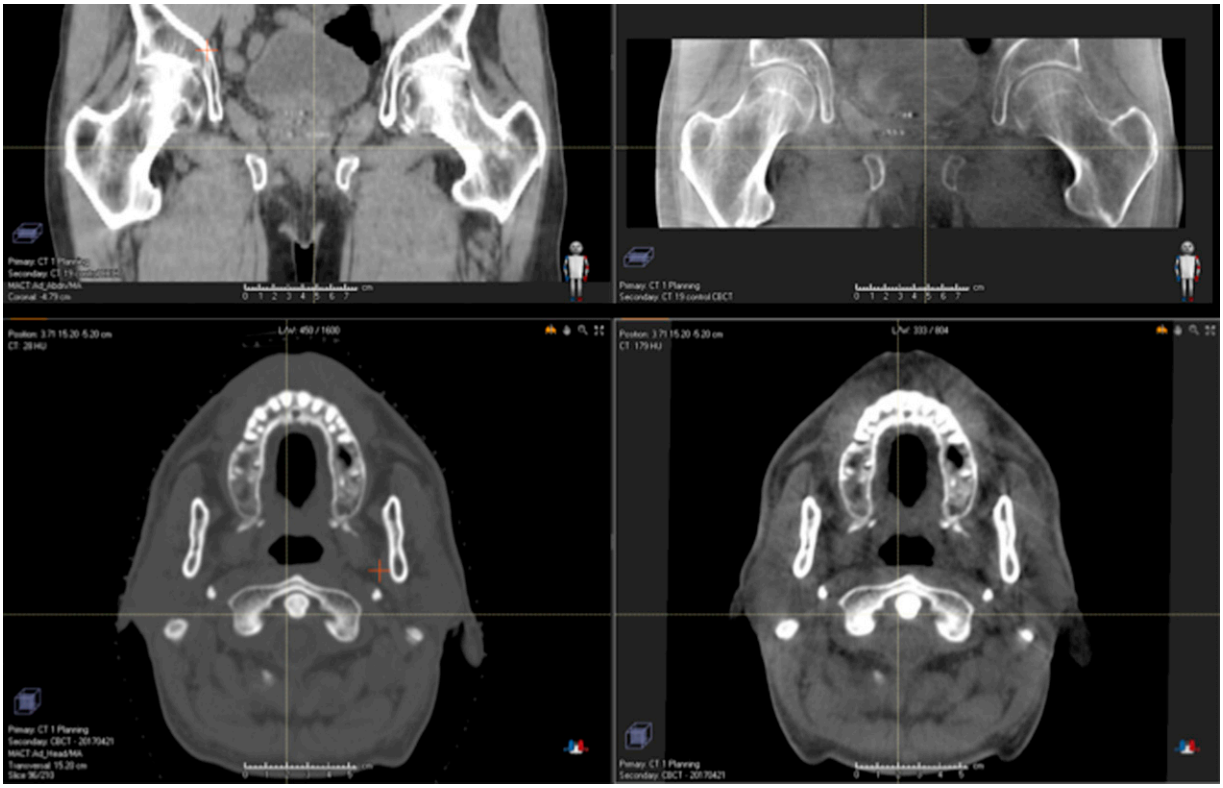

Figure 8. Planning CT for a prostate patient on the top left and CBCT generated by the ImagingRing™ in the treatment room on the top right. Planning CT for a head and neck patient on the lower left and CBCT on the lower right.

that we will soon clinically commission a Monte Carlo (MC) dose calculation algorithm implemented in RayStation v6.0. The algorithm is based on Fluka and a better performance not only in inhomogeneous media is expected [25]. As said, the pencil beam algorithm has some deficits in situations of large air gaps and usage of range shifter. With MC it is assumed that absolute dose calculation accuracy is improved and the necessity to rescale for treatment fields with range shifter is much more reduced [26,27].

On the other hand, reducing the air gap between patient and nozzle can potentially result in collisions of hardware components in the treatment room. Therefore, a dedicated software module, which is embedded in the TPS and in the Record & Verify software, was developed for MedAustron. The so-called imaging definition and collision avoidance system (IDCAS, medPhoton GmbH, Salzburg, Austria) allows checking for potential collisions of all components (immobilization equipment, treatment couch, kinematic model of the PAS, nozzle), including the patient (represented by an avatar and the patient surface, retrieved from the planning CT) in the treatment room. This step is performed prior a treatment plan is finalized. Fig. 11 shows the collision visualizer (software module of IDCAS), which is embedded in the TPS and by kinematic calculations based on the treatment plan a warning is generated about potential collision risks taking into account 3D safety margins. Hence, the air gap can be efficiently reduced without the risk of collision.

2.5 Fast, alternative patient specific QA procedures

One major bottleneck in the clinical implementation of adaptive radiotherapy with either photons or protons is linked to machine and patient specific quality assurance and the underlying legal basis for QA [26]. In general, for LIBT all plans applied to a patient need to be dosimetrically verified prior to treatment. This is very resource intensive on the machine side but also on the manpower side. Similar arguments have been communicated in the early days of clinical implementation of intensity modulated radiotherapy two decades ago. Similar to photon beam therapy, in LIBT the implementation of fast patient specific QA approaches like independent dose calculation [28], independent Monte Carlo plan verification or log-file analysis [29] is needed but rarely clinically implemented. At MedAustron a workflow which allows to use the log files to re-compute the dose for a specific fraction is implemented for research purposes (see Fig. 12). Software tools for a beam delivery evaluation based on a

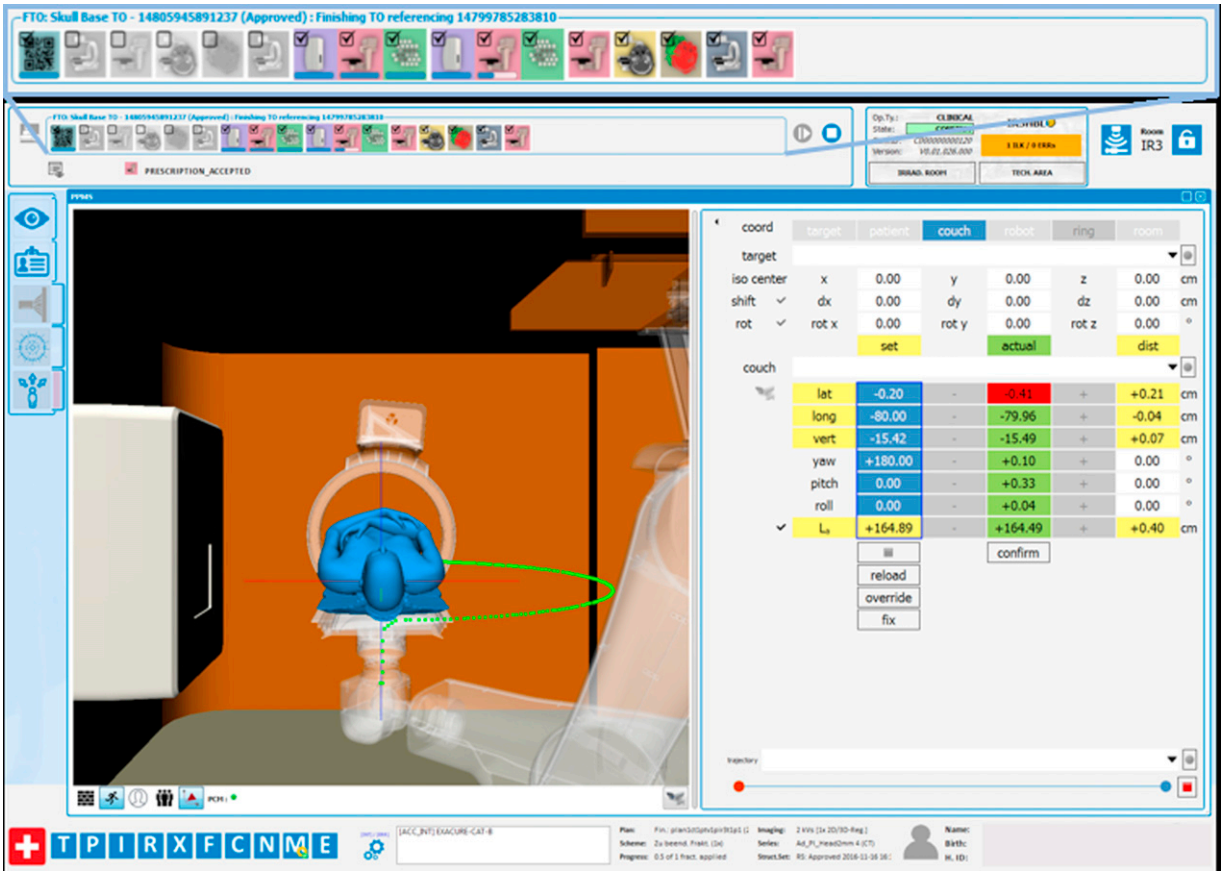

Figure 9. Screenshot of the RTS[2] software. The top row of symbols shows the processing of pre-planned workflow steps – so called treatment operations (zoomed out) – for a specific patient. This treatment operations can e.g. include robot movements, individualized 2D and 3D image acquisition, image registration, nozzle element movement, beam application and or barcode scanner based patient/immobilization device identification procedures.

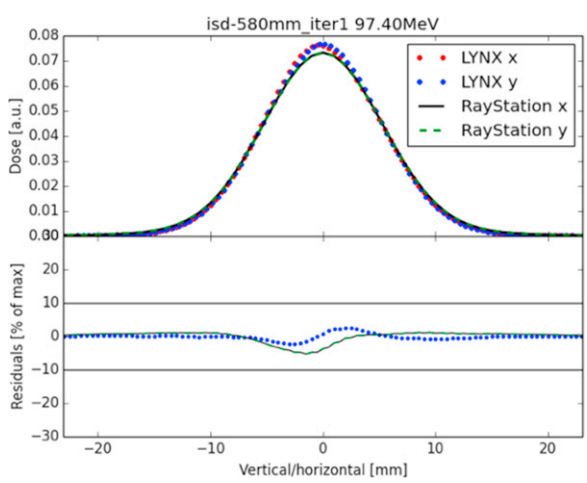

Figure 10. Comparison of spot profiles from the beam model and the Lynx measurements using a 7 cm air gap for a 97.4 MeV central pencil beam.

standardized treatment log-file analysis were developed that allow dose re-computation in the TPS based on log-file fluence distributions measured by the dose delivery system during delivery. For this purpose, DICOM RT ion plans are generated utilizing measured spot positions and spot weights which are then imported into the TPS. Re-computing dose for the delivered particle fluences allowed comparing delivered against planned dose distributions. Log-file analysis and dose re-computation has been performed for an initial test set of patient cases.

Initial tests revealed that deviations of fluences tend to lead primarily to dose deviations in the penumbra region and occur mostly for low-weighted spots. All evaluated treatment fractions from patients treated in the first 6 months of clinical operation show average dose differences of less than ±1% and gamma pass rates (3%/3 mm) above 95%. Quantities obtained from standard log-file analysis (Fig. 13) correlate with dose differences obtained from plan re-calculation and allow verifying precise plan delivery without the necessity

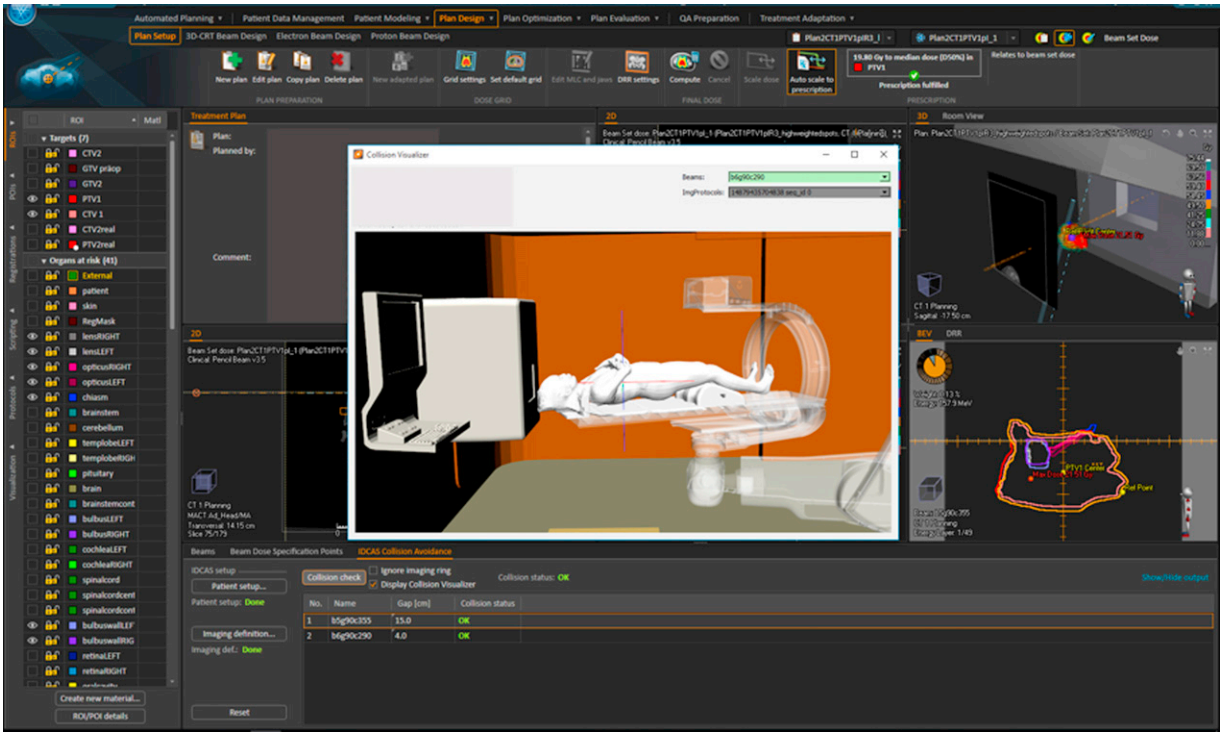

Figure 11. IDCAS collision visualizer embedded in the TPS to ensure that the user is warned already during treatment planning phase about potential patient and/or device collisions in non-isocentric or non-coplanar treatment setups.

of a full log-file-based dose re-computation. Fig. 13 shows distributions of deviations in spot positions and spot weights for a single fraction. Tolerance limits can be established for such quantities from log file analysis to assure dosimetrically accurate plan delivery.

3 Towards adaptive therapy

The previous section summarized the technical description of systems and concepts involved and highlighted key components for future adaptive LIBT treatment at MedAustron, while more detailed information on commissioning results of the respective components will be summarized in publications which are currently under preparation. Although adaptive LIBT is not yet clinical routine it can be concluded that the current technology provides the functionality for a variety of indications, which will benefit from adaptive LIBT.

At present commissioning at MedAustron is ongoing and addresses the following functionality extensions: clinical implementation of the vertical beam line and proton gantry, carbon ion beams, higher intensity beams, different spill lengths, dual-energy and fluoroscopic in-room imaging, deformable registration, scripted treatment planning. The following paragraphs will focus on further concepts of adaptive LIBT which can be addressed at MedAustron in the near future. X-ray and MR image guidance as well as in vivo range verification and automated planning are discussed in the context of general developments in radiotherapy.

3.1 X-ray image guidance

The fluoroscopy function of the ImagingRing™ will be a proper tool for the dynamic surveillance of tumors located in the thorax or abdomen region. Either during free breathing or during gated deep inspiration breath hold low frequency fluoroscopy (1 Hz to decrease the imaging dose) can be a used for intrafraction motion management. Currently, research is conducted to integrate ultra-fast 2D/3D online registration [30,31] to be able to automatically detect the tumor position instantaneously when the image is generated and to support markerless motion management during treatment with a rescanning technique. It needs to be stressed that there are possible ImagingRing™ versus beam line geometry combinations, where direct online imaging due to interference of for example the X-ray source with the vertical beam line is challenging. Here the integration of the collision avoidance system in the treatment planning process and the flexibility of the ImagingRing™ in selecting non-opposing X-ray source to detector views can be of great help to realize intrafraction imaging.

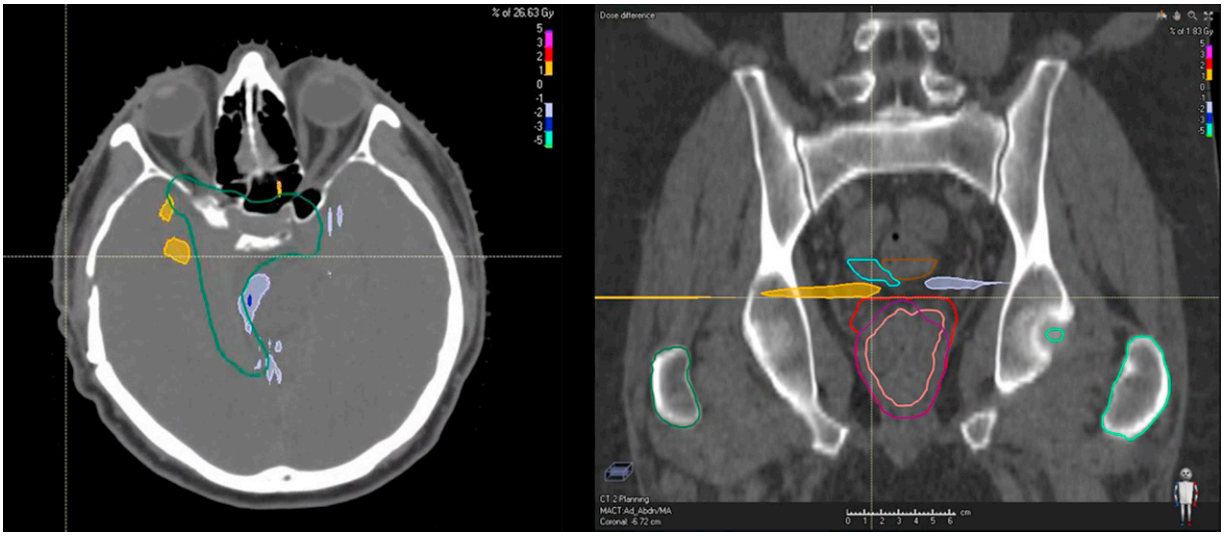

Figure 12. Dose differences in percent between the planned and the delivered dose distributions for a meningioma (left) and a prostate (right) patient for a single fraction.

The first clinical CBCTs are very promising and possibly enable to achieve more reliable results during deformable image registration as they are more similar to the planning CTs. Their potential in terms of image quality and also dose recalculation still needs to be investigated.

CBCT in conjunction with deformable image registration on planning CT can also be used to calculate water equivalent path lengths to trigger a treatment plan adaptation [32,33]. Several groups also demonstrated the possibility to use the daily in-room CBCT for deforming the planning CT and base the adaptive dose calculation on this virtual CT [34,35]. Several limitations still need to be addressed as e.g. limited FoV of the CBCT, validation of the deformable image registration. Finally, dual-energy CBCT could also potentially improve dose calculation accuracy for LIBT delivery once derivation of relative electron densities and stopping power ratios are proven to be applicable in adaptive workflows [3].

3.2 MR image guidance

MR images have a beneficial soft tissue contrast compared to CT and are well established for structure segmentation in radiation oncology including multi-parametric imaging for tissue characterization [36]. More recently MR based treatment planning, MR based response assessment [37] as well as novel IGRT technology based on hybrid MR linacs is investigated in photon beam therapy [38]. Solutions for converting MR images to synthetic CTs [39] have been proposed and are already commercially available (Siversson et al., 2015 [40]).

MR guided proton therapy has therefore become a recent research topic [38,41]. MR based proton dose calculation seems feasible [42]. For head and pelvic cases the heterogeneous synthetic CTs showed dose differences between 0.6% and 1.4% in the CTVs only compared to actual planning CTs. A further advantage of MR-only based planning is that registration errors between CT and MR are not relevant anymore. At MedAustron a 3T MRI is installed for treatment planning (currently MR-CT registration) and follow up response assessment. It is envisaged to investigate the Magnetic Resonance for Calculating ATtenuation technology (MR-CAT) and other comparable approaches developed for photon therapy and its potential use for the MR-only proton treatment planning workflow. For that reason, all immobilization equipment used is MR-compatible. In case MR-only planning will be implemented, weekly control-MRs can then be used for treatment plan adaptation without the need of additional ionizing radiation. Besides converting MR images into synthetic stopping powers for dose calculation the development of dose calculation engines for protons in the presence of magnetic fields is ongoing [43].

3.3 In vivo range verification

The double-sided sword of ion beams is the sharp distal fall off to better spare organs at risk but on the other hand the precision is highly influenced by range uncertainties. PET based range verification seems to be an "evergreen" in LIBT research and certainly has the potential to trigger treatment adaptations in LIBT [44]. Motivated by the encouraging results obtained at the Gesellschaft für Schwerionenforschung (GSI)

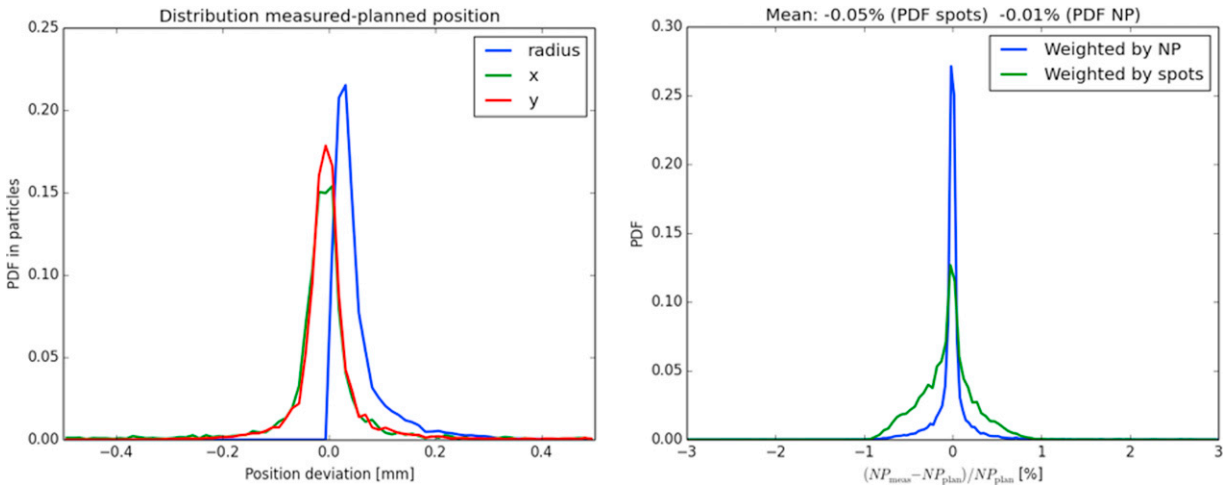

Figure 13. Probability distribution functions (PDF) of deviations in spot positions (left) and spot weights (right) for a single fraction are shown for an exemplary patient.

and based on the in-room camera developed at Rossendorf, several groups investigated this approach. The feasibility and first clinical results of offline PET range verification was shown by a group at HIT [45,46] and by a group at Kashiwa [47]. However, PET inherent challenges such as the limited spatial resolution and metabolic factors of influence and the lack of commercial and dedicated solutions have hampered progress in its broad clinical implementation. Although in-beam PET as e.g. further investigated by a group at MD Anderson [48] shows higher activity and more reliable results, today mostly offline solutions are pursued. For example, the group at HIT demonstrated that offline PET images still contain valuable information for treatment verification [45]. At MedAustron a similar approach using offline PET range verification will be explored to detect necrotic tissue and hence to detect potential tumor biology changes during treatment. A patient transfer system based on a shuttle with hoovering technology minimizes the time between treatment and PET acquisition and keeps the patient immobilized.

For protons, PET based range verification has been superseded by prompt gamma detection methods. This approach has resulted in a commercial tool and its clinical potential to extract 1D information has been recently demonstrated [49]. A unique feature and advantage of prompt gamma detection is that real time information during treatment can be generated and hence better clinical conclusions can be drawn in case a change in range is detected as it is not influenced by decay or washout. Prompt gamma measurements are better correlated with the Bragg Peak position and are also promising for carbon ions. Clinical introduction of 1D analysis was shown, but further work needs to be done to extract 2D or even 3D information. Further the size of e.g. used slit cameras are small compared to PET scanners and hence with a proper trolley or even a robot-mounted setup all body regions can be imaged in fixed beam lines as well as gantry rooms. Nevertheless, careful integration into the collision avoidance and detection system would be needed as the system would increase the complexity in the room modelling.

3.4 Automated planning

In photon therapy adaptive treatment planning has not yet reached widespread clinical implementation, mainly because it is considered to be resource intensive in many aspects. One dominant aspect is the time-consuming task of re-planning. Although there is the possibility to prepare templates and/or plan libraries for specific situations and treatment sites, respectively, still manual tailoring of the treatment plan is often needed to fulfill clinical constraints for every single patient. Consequently, plan quality might still vary depending on the experience of the planner. As soon as regular plan adaptation is needed the limited resources become a bottleneck similar to patient specific quality assurance in the adaptive process. Automated plan generation would therefore not only speed up this process but also increase the consistency in the plan generation. In photon therapy substantial progress was made in the last years [50,51]. For example Sharfo et al. [51] showed that plan quality was comparable or better and planning time dramatically decreased. Considering that about one third of all LIBT plans need treatment adaptation, automated planning is even more important for LIBT. A recent study showed the feasibility of automated proton plan adaptations based on CBCT images [52]. The envisioned process at MedAustron will be based on high quality in-room CBCT images and the use of deformable registration to generate a virtual CT from the initial planning CT including drawn

structures. Such a procedure can save some time for contouring but still needs check and adaptation of contours and final approval by a physician. As prerequisite the uncertainty of the deformable image registration algorithm has also to be defined [53].

Finally, scripting functionality of state-of-the-art TPS can support to fast generate clinical plans on this virtual CT by applying templates. In the long run, it is envisioned to use the dual-energy CBCT functionality to facilitate dose calculation on the CBCT images itself. This might not be possible in all situations as the FoV of the CBCT e.g., in head and neck patients, might not cover all necessary regions.

4 Conclusion

In conclusion, in-room imaging and all its purpose built auxiliary equipment at MedAustron already allows for high quality proton therapy and represents the technological basis for future adaptive concepts. Most important steps towards clinical realization of adaptive LIBT will be the commissioning of the MC dose calculation algorithm in the TPS, fluoroscopic imaging, dual-energy CBCT and alternative and fast patient specific QA methods will be implemented in the coming months and years. Advanced research activities will be pursued in future towards clinical implementation of MR-based planning, as well as automated planning and in-vivo range verification methods.

Conflict of interest

No conflict of interest for any of the authors.

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